



THE REINCARNATION OF LOGISTICS

ADAPTIVE REUSE FOR URBAN LAST-MILE DISTRIBUTION

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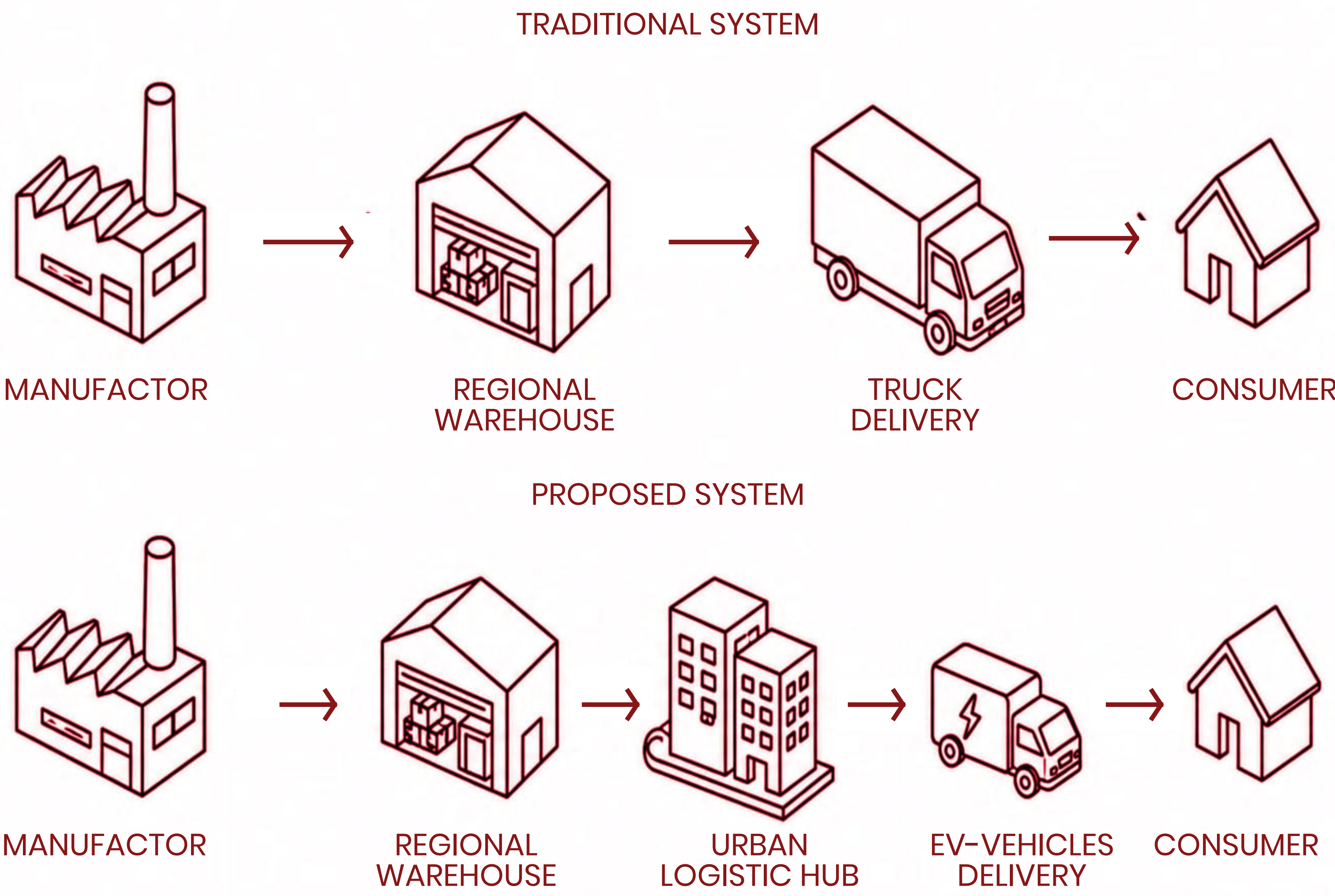


STUDIO BRIEF

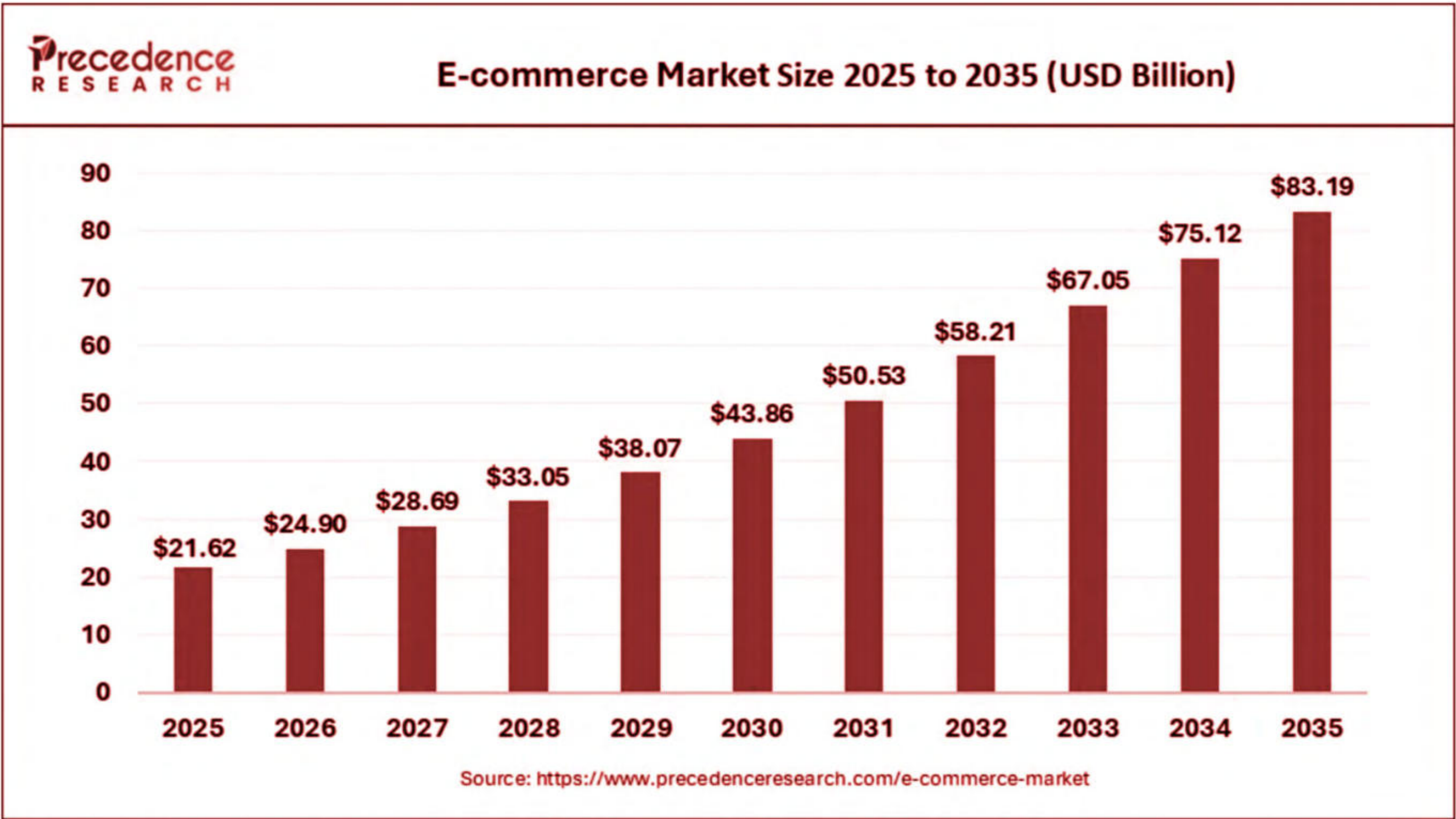
The studio "Rethinking Logistics" explores how logistics shapes contemporary architecture and urban environments. Through lectures, research, and analytical studies, the studio examines logistics systems beyond warehouses, including data centres, manufacturing, distribution networks, last-mile delivery, and micro-hubs.

The research highlighted challenges such as congestion, inefficiency, and the separation of logistics infrastructure from the city, while also exploring how architectural and urban interventions can address these issues. A key part of the studio is the use of KPIs to evaluate energy performance, spatial efficiency, environmental impact, and urban integration.

The brief questions the conventional model of isolated warehouses outside cities and instead investigates how logistics can be re-integrated into urban areas through adaptive reuse and more sustainable design strategies. The project aims to develop architectural solutions that combine logistics systems with public urban life in a more accessible and integrated way.



E-COMMERCE GROWTH



The large increase of e-commerce has significantly increased the demand for fast and efficient last-mile delivery systems within cities. As online shopping continues to expand, logistics infrastructure has become an essential part of everyday urban life. However, traditional logistics hubs are often located outside the city, creating longer delivery distances, increased traffic, and higher carbon emissions.

With the growth of e-commerce, logistics systems are under increasing pressure to operate faster and more efficiently within urban environments. As distribution networks continue to expand across large distances, challenges related to delivery times, operational efficiency, and urban congestion become more visible. The project explores how logistics infrastructure can be more closely integrated into the city through localized distribution and adaptive reuse strategies.

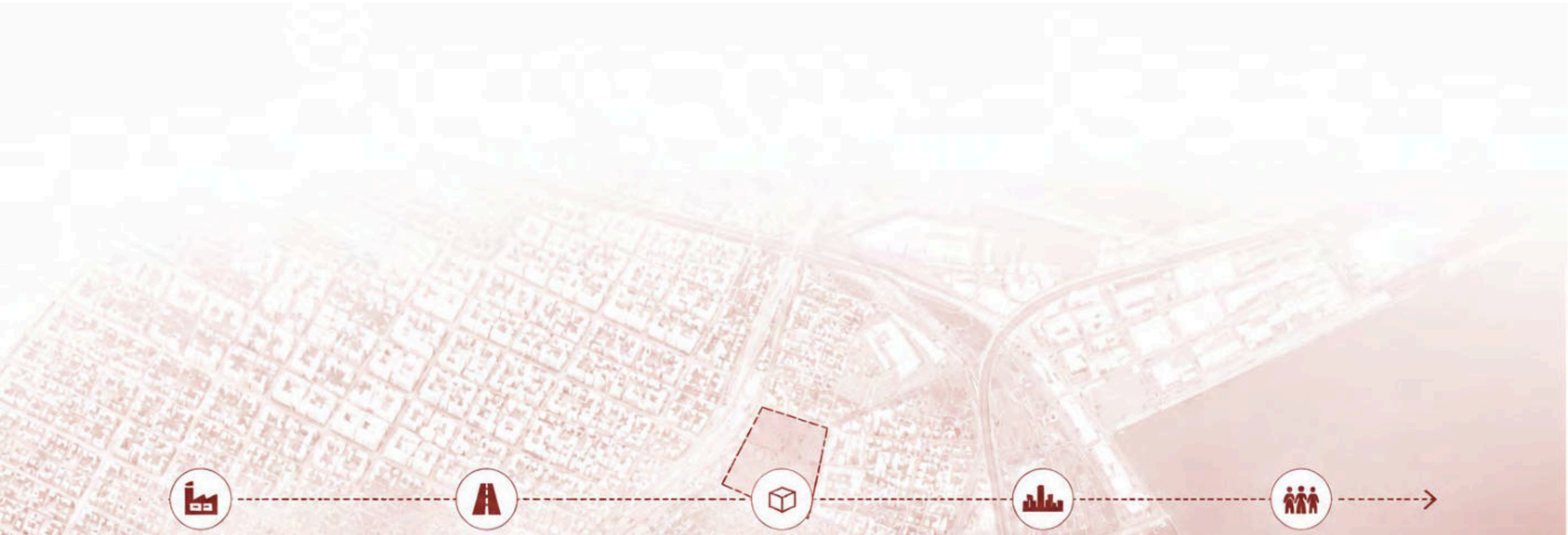


CONTEXT

The rapid growth of e-commerce is placing increasing pressure on urban logistics systems, particularly within last-mile delivery. Traditional logistics facilities are often located outside cities, creating long transport distances, congestion, and high emissions. As demand for faster delivery continues to increase, logistics infrastructure must move closer to the end user.

At the same time, many post-industrial districts contain abandoned or underutilized buildings that are frequently demolished despite their existing structural value. Adaptive reuse offers an opportunity to reduce embodied carbon while reactivating existing urban infrastructure.

The project is located in a transforming industrial district in Prague, where former production areas coexist with new residential developments and transport infrastructure. The proposal explores how an abandoned industrial building can be transformed into an urban logistics hub that integrates public space, environmental strategies, and efficient last-mile distribution.



PROBLEMS



In recent years, the big change of global e-commerce has significantly transformed logistics systems. Market forecasts indicate that the global e-commerce sector is expected to expand from approximately 21.62 trillion USD in 2025 to 83.19 trillion USD by 2035. This exponential growth places increasing pressure on last-mile delivery, which is already one of the most complex, costly, and carbon-intensive stages of the supply chain.

To accommodate this demand, logistics infrastructure must move closer to the end user. One emerging strategy is the integration of last-mile distribution hubs within urban environments, enabling shorter delivery distances and more efficient use of low-emission transport such as cargo bikes and electric vehicles. However, dense cities face a critical limitation: lack of available space.

At the same time, many urban areas contain a significant number of abandoned or underutilized buildings, often remnants of post-industrial development. These structures are frequently demolished to make way for new construction, despite the environmental cost. Demolition and new construction result in substantially higher embodied carbon emissions compared to adaptive reuse.

Using a simplified embodied carbon calculation ($\text{area} \times \text{kgCO}_2\text{e/m}^2$), renovation typically results in approximately 250 $\text{kgCO}_2\text{e/m}^2$, while demolition and new construction can reach around 650 $\text{kgCO}_2\text{e/m}^2$. This highlights a significant opportunity: reusing existing buildings as urban logistics hubs can simultaneously address spatial constraints and reduce carbon emissions.

LOCATION

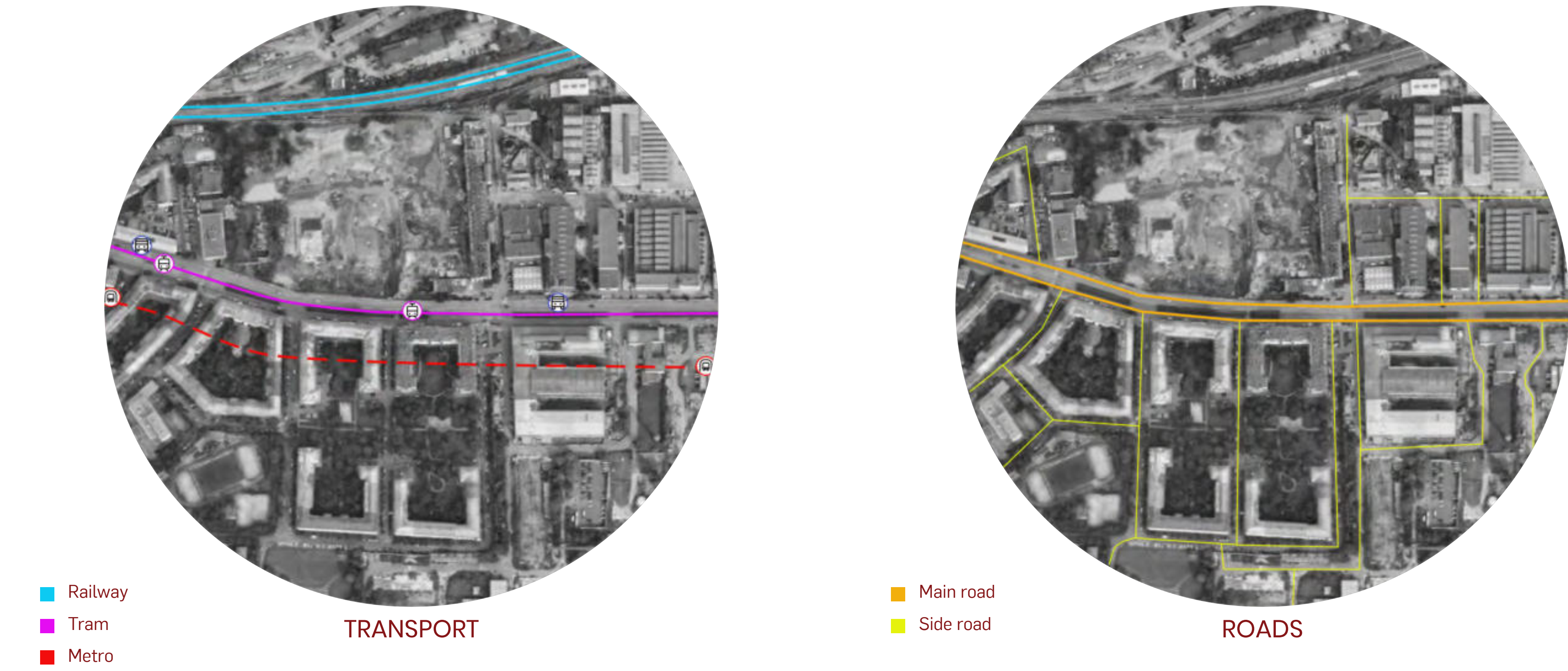
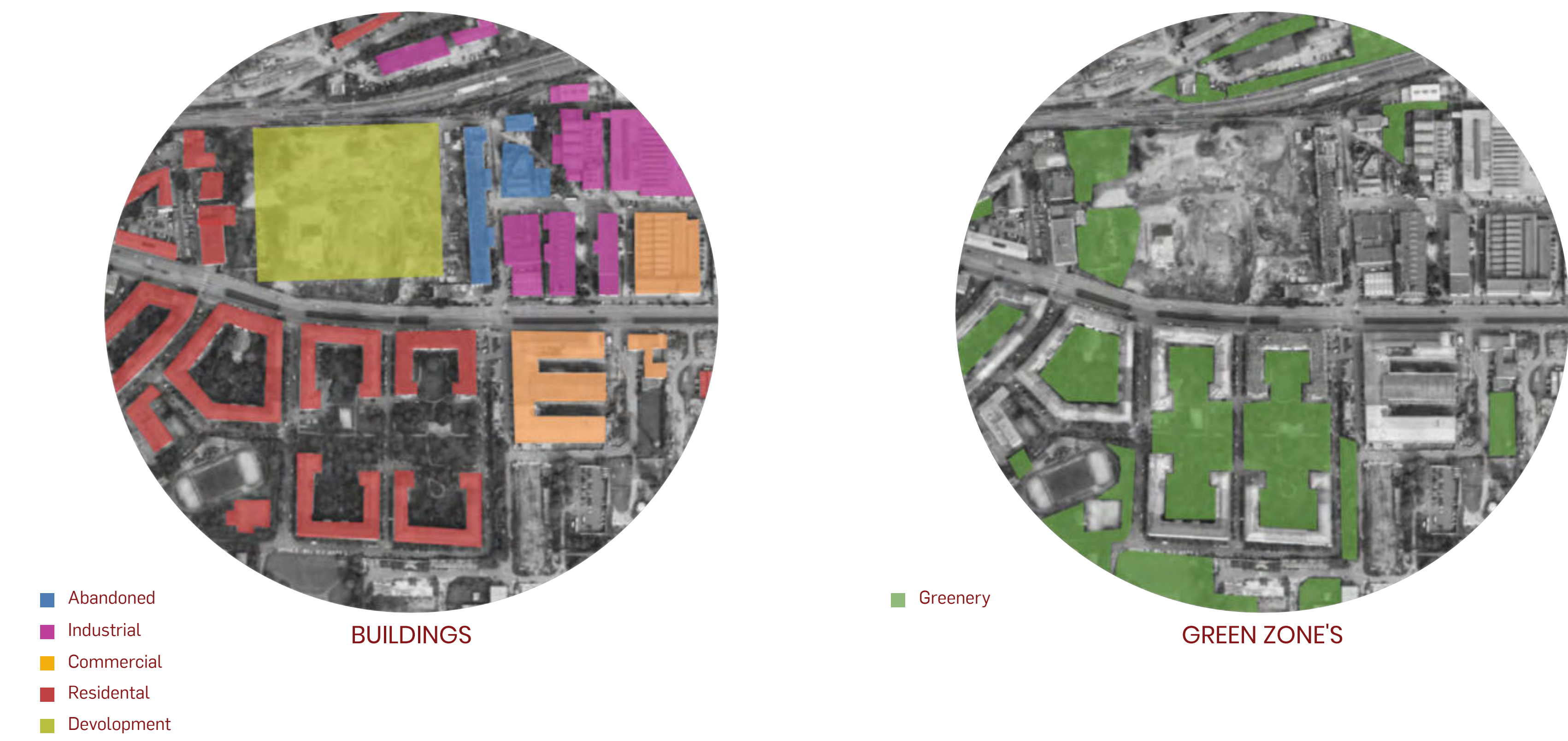


The project is located in a post-industrial district of Prague 9, where former industrial buildings coexist with emerging residential developments. This transition reflects a broader urban shift, as previously production-focused areas are gradually reintegrated into the contemporary city.

The site consists of an underutilized industrial structure with strong connections to surrounding transport infrastructure. Its position within a growing urban environment creates an opportunity to redefine its role, transforming it into an active part of the city while responding to both logistical and environmental challenges.



SITE ANALYSIS



CASE STUDY



A relevant example of compact urban logistics is the DPD micro-hub in Westminster. The facility occupies approximately 460 m² and operates as a small-scale last-mile distribution center, where parcels are transferred from larger delivery trucks to low-emission vehicles such as electric vans and cargo bikes. This significantly reduces traffic congestion and emissions in dense urban areas while maintaining high delivery efficiency.



A similar strategy is implemented in France, where a micro-hub is located beneath a highway. Although slightly less documented, the facility follows the same principle of utilizing residual urban spaces for logistics. By activating otherwise unused infrastructure zones, these hubs demonstrate how small-scale interventions can support growing e-commerce demands without requiring large new developments.

CONCEPT

The project proposes the adaptive reuse of an abandoned industrial structure as an urban last-mile logistics hub. Instead of demolishing the existing building, the proposal transforms it into a hybrid infrastructure that combines logistics operations, public space, and environmental strategies within a dense urban context.

The concept is based on a dual-sided approach. One side of the building is dedicated to logistics functions, including truck access, loading docks, storage, and outbound electric delivery vehicles. The opposite side engages the public area through the introduction of a landscaped plaza with vegetation, water elements, and seating areas.

By separating industrial operations from public activity, the project creates a balance between efficiency and accessibility. The proposal challenges the conventional image of logistics architecture as isolated and inaccessible, instead redefining it as an integrated part of the community.

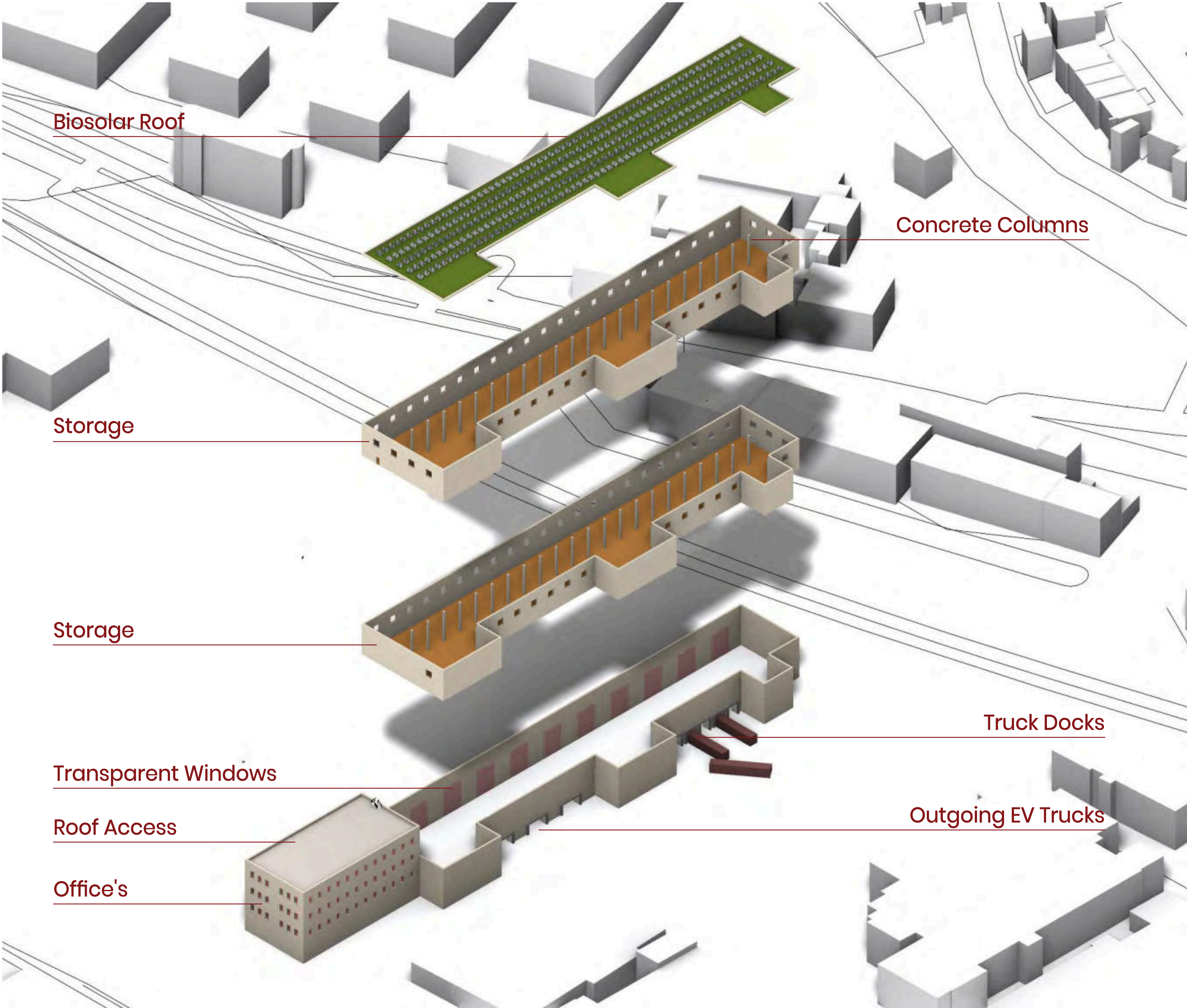


SITE MAP

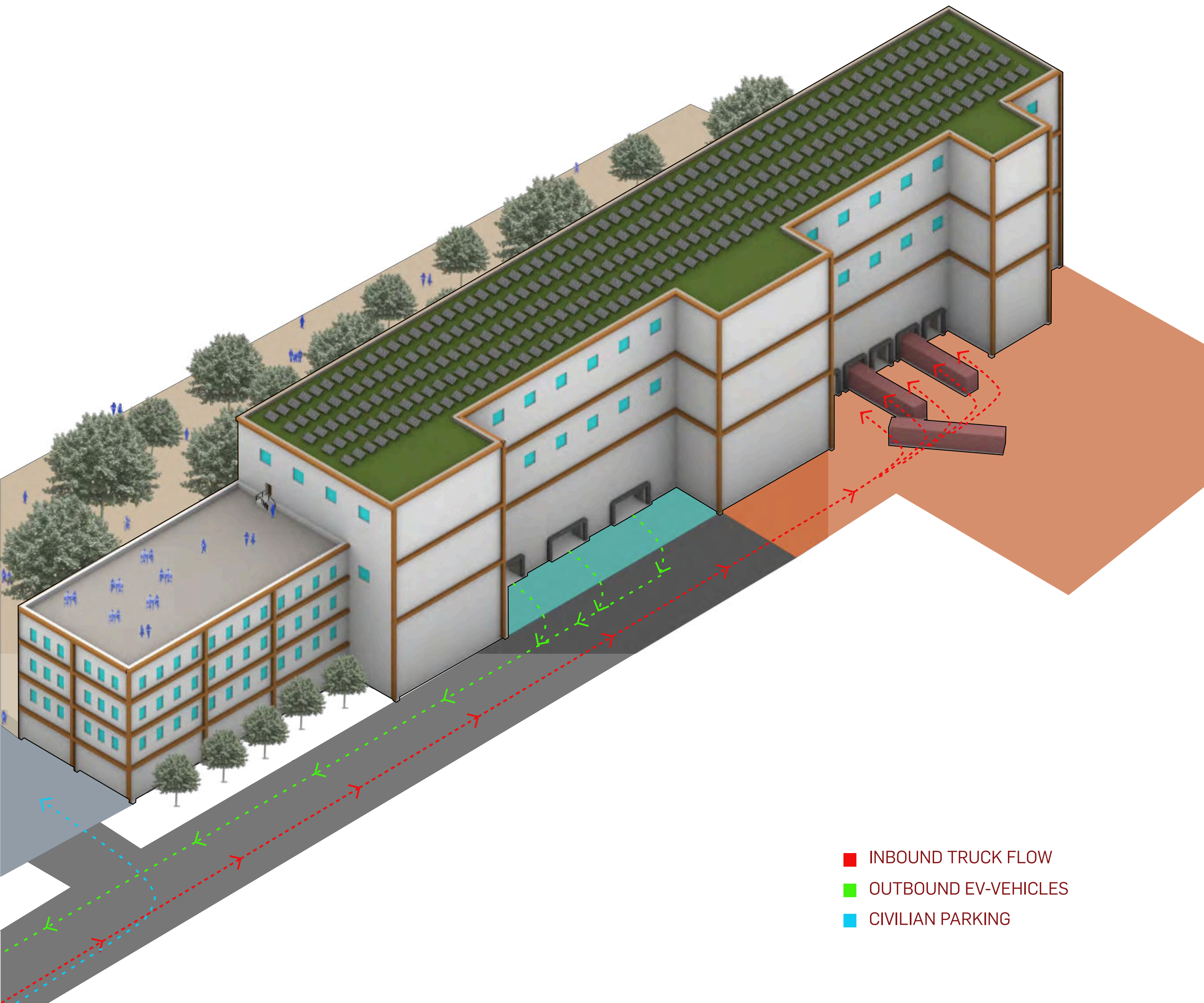


Truck Zone EV-Vehicals Zone Public Plaza Parking Vehical's Road

BUILDING STRUCTURE



FLOW OF VEHICLES



PUBLIC ACCESS

Logistic hubs are typically closed environments, designed solely for operational efficiency with little consideration for public engagement. As a result, they often appear uninviting and inaccessible, discouraging people from approaching them due to concerns related to safety and aesthetics.

This project challenges that condition by introducing a dual-sided approach. One side of the building is dedicated entirely to logistics operations, accommodating truck movement, loading, and last-mile distribution in a controlled and efficient manner.

In contrast, the opposite side is designed to engage the public. A plaza is introduced as an interface between the building and its urban context, incorporating water elements, green spaces, and seating areas. This transforms the perception of the logistics hub from a closed industrial facility into an part of the neighborhood.

By creating an inviting public space, the project encourages nearby residents to spend time in the area, fostering interaction with both the building and its surroundings while maintaining a clear separation between public and industrial functions.





PERFORMANCE ANALYSIS

This section explores the environmental and urban strategies that inform the proposal. Through research on biosolar roofs, electric delivery systems, and embodied carbon, the project investigates how logistics infrastructure can become more sustainable and better integrated within the urban environment.

Key Performance Indicators (KPIs) are used throughout the project to evaluate the impact of these strategies. By combining qualitative design decisions with measurable data, the proposal is assessed in terms of energy production, carbon reduction, public accessibility, and environmental performance.



KEY PREFORMANCE INDICATOR'S

Delivery Emission's

The integration of electric delivery vehicles reduces operational emissions compared to conventional combustion-engine logistics vehicles operating within the urban environment.



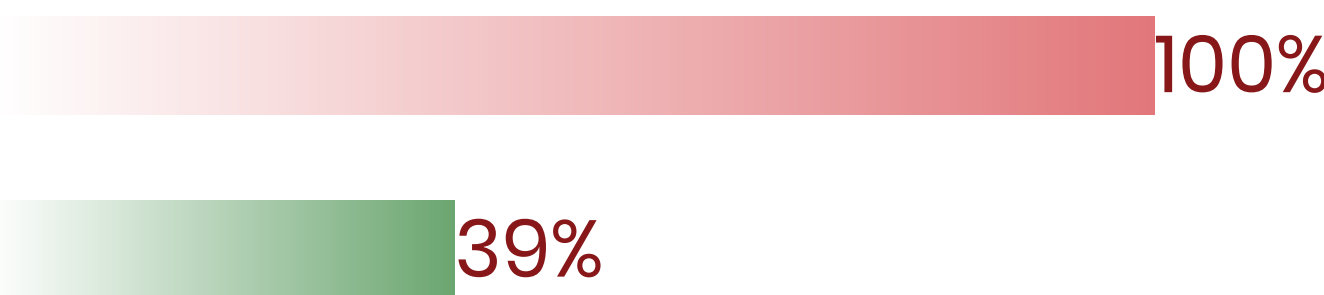
Public Accessibility

The proposal introduces programmed public space, green infrastructure, and clearer pedestrian integration with the surrounding neighborhood.



Embodied Carbon

Emodied carbon is a big problem in architecture. And with abandoned buildings and rebuilding the number gets even bigger. So the proposal is insted of demolish to renovate to lower the carbon footprint of the building



ROOF UTILIZATION

Since the building was abandoned the roof was not in use. There do we calculate the percentage of the roof that was used when the building was active. and compare to what it will be.



BIOSOLAR ROOF



ROOF USAGE
BEFORE



ROOF USAGE
AFTER



The biosolar roof combines renewable energy production with ecological performance by integrating photovoltaic panels with vegetated roof surfaces. This strategy increases on-site solar energy generation while improving biodiversity through habitats for insects and native vegetation, contributing to stormwater retention, thermal regulation, and a more sustainable urban environment.

EV-VEHICLES



COMBUST
ENGINE
VEHICLES

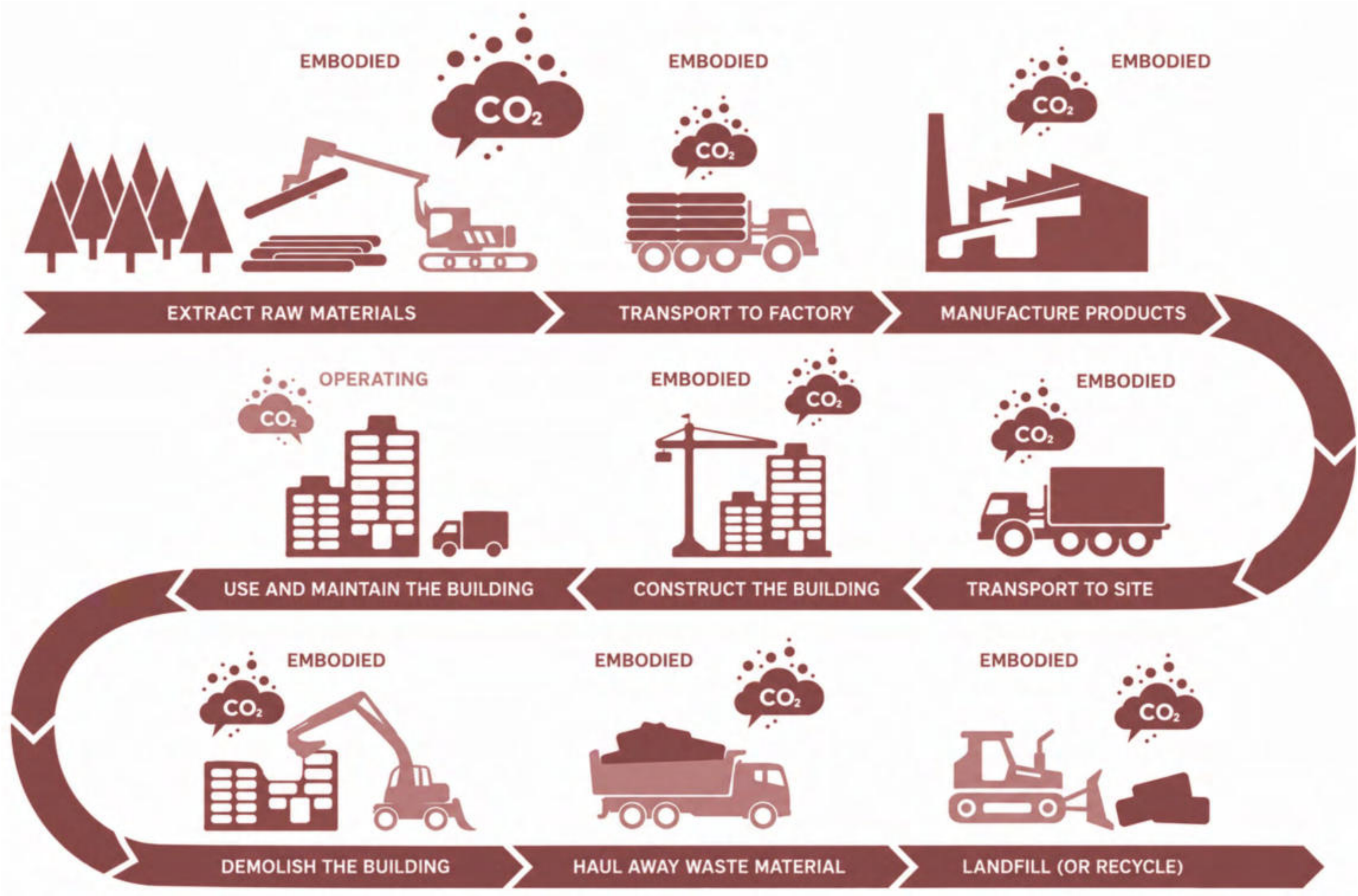


EV-VEHICLES



The implementation of EV delivery vehicles reduces carbon emissions compared to traditional combustion-engine transport. By supporting cleaner last-mile distribution, the project lowers operational emissions, reduces noise pollution, and contributes to a more sustainable and environmentally integrated logistics system.

CARBON EMISSION'S



DEMOLISH + REBUILD 1,566 tCO₂e

RENOVATION 450 tCO₂e

The adaptive reuse of the existing industrial structure significantly reduces embodied carbon emissions compared to demolition and new construction. Instead of replacing the building, the proposal preserves the existing structural framework and integrates new environmental systems, reducing material consumption and construction-related emissions.

By reusing the existing building, the project minimizes waste production while extending the life cycle of the structure. The integration of EV-based last-mile delivery and biosolar roof systems further contributes to lowering the environmental impact of the logistics operation. The project demonstrates how urban logistics infrastructure can become both environmentally responsive and spatially integrated within the contemporary city.

PUBLIC PLAZA



PUBLIC ACCESS BEFORE 3%

PUBLIC ACCESS AFTER 90%

The introduction of a new public plaza transforms the previously inaccessible industrial site into an active urban space that encourages pedestrian movement, social interaction, and community engagement. By reallocating underused logistics areas into publicly accessible green space, the proposal improves urban integration while creating a safer and more welcoming environment for both residents and visitors.

REFLECTION

The project began with a question of how logistics infrastructure could exist within the city in a more integrated and sustainable way. Rather than treating logistics as hidden industrial architecture placed outside urban areas, the proposal explores how these systems can become part of everyday urban life.

Through the adaptive reuse of an existing industrial building, the project investigates the relationship between logistics, public space, and environmental performance. Strategies such as EV-based delivery, biosolar roofing, and the reduction of embodied carbon were used to rethink how a logistics hub can operate while contributing positively to its surroundings.

The process highlighted the growing impact of e-commerce and last-mile delivery on contemporary cities, while also demonstrating the potential for architecture to improve these systems through design. By combining research, spatial interventions, and KPI-based analysis, the project proposes a logistics hub that is not only functional, but also more accessible, visible, and connected to the urban environment.

Ultimately, the project suggests that future logistics infrastructure does not need to remain separated from the city, but can instead become an active and integrated part of it.



